

UTSAV CHAUDHARY

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Senior **DevOps Engineer** with **5+ years** supporting and delivering **production-grade AWS and Azure infrastructure** for high availability systems in regulated and enterprise environments. Proficient in **Kubernetes and Helm orchestration**, **Infrastructure as Code** with **Terraform** and **Ansible**, and **multicloud deployments** across Dev, Test, UAT, QA, and Production. Experienced contributing to **disaster recovery**, **observability**, and compliance-ready platform operations that meet **RTO** and **RPO** targets for business-critical systems.

Technologies

Cloud & Infrastructure: AWS (EC2, EKS, ECS, Fargate, Lambda, RDS, S3, EBS, VPC, IAM, CloudWatch, API Gateway, Secrets Manager, Route 53, ALB/ELB, CodeDeploy, KMS, CloudHSM), Azure (App Services, AKS, Azure DevOps, Azure SQL, Application Gateway, Front Door, Key Vault, Entra ID), Docker, Kubernetes, Helm, Rancher, Argo CD

IaC & Automation: Terraform, Ansible, Puppet, Chef, Bash, Python, PowerShell, YAML, CloudFormation, Nginx, Cron

CI/CD & Version Control: Jenkins, GitHub Actions, Bitbucket Pipelines, AWS CodePipeline, Azure DevOps, Git, GitHub, GitLab, Bitbucket, AWS CodeArtifact

Monitoring & Logging: Prometheus, Grafana, Elasticsearch, Logstash, Kibana, CloudWatch, Application Insights, Datadog, Elastic Stack

Networking & Security: VPCs, VNets, Security Groups, NSGs, IAM Policies, DNS, VPN, LDAP, RBAC, SSL/TLS, OAuth 2.0, HIPAA, GxP

Development & Tools: Python, TypeScript, JavaScript, Groovy, FastAPI, Node.js, React.js, Next.js, REST APIs, gRPC, WebSockets, PostgreSQL, MongoDB, Redis, Firebase, Kafka, Jira, Confluence

Operating Systems: Linux (Ubuntu, CentOS, RHEL), Windows Server, macOS, Raspberry Pi

Experience

Sr. DevOps Engineer, Hybrid, **Labs196**, TX

Feb 2025 - Present

- Collaborated with platform and compliance teams to support a **multicloud strategy** across **AWS** and **Azure**, helping deploy and maintain **Kubernetes** workloads on **EKS** and **AKS** with **Helm** across **Dev**, **Test**, **UAT**, **QA**, and **Production** for regulated lab and research platform services.
- Contributed to **GitOps** release workflows using **Argo CD**, collaborating with platform and staff engineers to standardize application deployments and reduce manual release steps across **GxP-validated** environments.
- Supported centralized cluster operations with **Rancher** managing **EKS** and **AKS** clusters, integrating **Azure Entra ID** for **RBAC**, least-privilege access, and audit-ready logging aligned with **FDA** and internal compliance reviews.
- Maintained multicloud infrastructure through **Terraform** and **Ansible**, managing **AWS** resources including **VPCs**, **security groups**, **IAM policies**, and **RDS**, and **Azure** resources including **VNets**, **NSGs**, **Key Vault**, and **Azure SQL** with version-controlled **IaC**.
- Contributed to cross-cloud **observability** using **Prometheus**, **CloudWatch Logs**, and the **ELK Stack**, supporting on-call incident response and production **SLA** targets for platform and data pipeline services.
- Helped automate backup and retention workflows for **S3**, **Azure Blob Storage**, and cloud-hosted databases, following **GxP** retention requirements established by compliance and platform leadership.
- Participated in **on-call** rotations and **incident response**, creating **runbooks** and coordinating with engineering teams to troubleshoot and restore production services across both cloud environments.

DevOps Engineer, On-Site, **Delta Dental**, GA

Jan 2024 - Feb 2025

- Collaborated with a distributed platform team, including **California-based engineers**, to support production **Azure Kubernetes Service (AKS)** clusters running **containerized claims and payor workloads** behind **Azure Application Gateway**, helping maintain **99.9% availability** for **member eligibility** and **claims processing** services.
- Contributed to centralized **monitoring and alerting** using **Application Insights**, **Azure Monitor**, and **Grafana dashboards**, integrated with **Azure Front Door** health probes and gateway metrics, supporting a team reduction in **mean time to detect (MTTD)** from **30 minutes** to **5 minutes** across **claims adjudication** and **enrollment** workflows.
- Assisted in migrating legacy applications to **cloud-native AKS** on Azure, working with senior DevOps engineers to improve **resource utilization** by **40%** for the **member enrollment** and **provider portal** platforms.
- Supported **canary release** rollouts through **Azure DevOps** pipelines with **Application Gateway** traffic splitting under senior guidance, contributing to a **75%** reduction in deployment-related incidents for the **member self-service portal**.
- Helped automate **backup** and **disaster recovery (DR)** for **Azure SQL** and cloud-hosted data services, supporting team targets of **RPO 10 minutes** and **RTO 30 minutes** for critical **claims transaction processing** systems.

- Supported **HIPAA-aligned** security practices using **Azure Key Vault** for encryption and key rotation of **member PHI and claims data** at rest and in transit, following standards set by senior engineers and the compliance team.

DevOps Engineer CO-OP, On-Site, **Midh Technologies**, TX

June 2023 - Dec 2023

- Managed production infrastructure for ML workloads, supporting **GPU backed Kubernetes clusters** on **AWS EKS** with cost optimization strategies including node pool configuration, resource quotas, and auto scaling policies.
- Designed high-performance inference infrastructure deploying **FastAPI** endpoints behind **AWS API Gateway** with **IAM authentication**, request throttling, and **Lambda** integrations for scalable ML inference.
- Developed infrastructure automation through **Ansible playbook** development, reducing manual server configuration and deployment time by **50%** while ensuring configuration consistency.
- Built production **CI/CD pipelines** with **GitHub Actions**, automating testing, Docker image builds, security scanning, and deployments to **AWS Amplify** and **EKS** environments.
- Implemented infrastructure security practices including **RBAC** policies, **IAM role** management, and **VPC** security configurations across AWS services and application layers.

DevOps/Platform Engineer, On-Site, **Compendious Medialabs Pvt. Ltd.**, IN

June 2021 - July 2022

- Managed hybrid infrastructure for **on premise Linux servers** (Ubuntu) with containerized deployments using **Docker**, establishing infrastructure reliability and automation through **Bash** scripting and configuration management.
- Designed and implemented self hosted CI/CD infrastructure deploying **GitHub Actions runners** on local infrastructure, building production grade pipelines for automated testing, Docker builds, and multiple environment deployments.
- Established production grade reverse proxy infrastructure with **Nginx** and **SSL/TLS** termination, optimizing external service exposure with **250ms** average response time and high availability.
- Implemented observability strategy deploying **Prometheus** and **Grafana** monitoring on local infrastructure; automated infrastructure tasks with **Python/Bash** scripting, reducing manual operational overhead by **60%**.

Software Engineer / DevOps, Remote, **VRoKCs**, MO

March 2020 - Sept 2020

- Managed production infrastructure on **Azure**, deploying **Flask web applications** on **Azure App Services** with high availability configurations, auto scaling, and integrated monitoring.
- Built infrastructure automation through **Azure DevOps** pipelines for ML model deployment, implementing artifact storage with **Azure Blob Storage** and version control workflows.
- Designed containerized infrastructure using **Azure Container Instances** and **Docker** for ML inference services, ensuring environment parity between development and production with consistent deployment practices.
- Developed CI/CD infrastructure using **Azure DevOps** for ML lifecycle management, automating training and deployment workflows and reducing deployment cycles by **40%** through pipeline optimization.

Web/Game Developer, On-Site, **underDogs Gaming Studios**, IN

March 2019 - Aug 2019

- Contributed to backend API infrastructure for gaming services, integrating and optimizing **Quarters API** across multiple games with secure transaction, supporting **load balancing** and **API rate limiting** for high availability systems.
- Supported infrastructure reliability initiatives by optimizing real time client server communication, validating and refining backend interaction flows using **Python** and **C#**, implementing error handling and health monitoring.
- Developed unified transaction infrastructure connecting external services (**payments, inventory, shipping**) to internal game systems, following consistent API patterns and infrastructure best practices.
- Participated in infrastructure security and testing practices, implementing **API security best practices**, comprehensive logging, and integration testing frameworks for edge cases including payment failures and network resilience.

Education

State University of New York at Binghamton

Master of Science in Computer Science

Aug 2022 - Dec 2023

3.44/4.00 GPA

Certifications



Microsoft Certified: DevOps Engineer Expert

AZ-400: Designing and Implementing Microsoft DevOps Solutions

AZ-900: Microsoft Certified: Azure Fundamentals

Google IT Automation with Python (Specialization)

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